

Practice Problems for Chapters 3&4

Chapter 3

1. Given a function $f(x)$,
 - (a) Find the slope of the tangent line to the graph of f at a general x -value by using $m_{tan} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ or $m_{tan} = \lim_{x_0 \rightarrow x} \frac{f(x_0) - f(x)}{x_0 - x}$.
 - (b) Find the equation of the tangent line to the graph of f at $x = 2$ using the result from (a).
 - (1) $f(x) = x^2 + 1$.
 - (2) $f(x) = 1/x$.
2. Find $f'(x)$ by using the definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ if
 - (1) $f(x) = x^3$.
 - (2) $f(x) = \frac{1}{\sqrt{x}}$.
3. Let f be a differentiable function with $f'(x) = \sqrt{3x+7}$, $g(x) = x^2 - 1$, and $F(x) = f(g(x))$. Find $F'(x)$.
4. Find $\frac{dy}{dx}$ if
 - (1) $y = x^{10} + 3x^{2/3} + \frac{4}{\sqrt{x}}$.
 - (2) $y = (x-1)^3 \sqrt{3x+1}$.
 - (3) $y = \left(\frac{1+x^2}{1-x^2} \right)^{17}$.
 - (4) $y = \tan(x^2 + 4x + 3)$.
 - (5) $y = x \cos x^2$.
 - (6) $y = \csc(\sqrt{x})$.
 - (7) $y = (1 + \sec x)(x^2 - \tan x)$.
5. Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ if
 - (1) $y = \frac{x-1}{x+1}$.
 - (2) $y = \cot x$.

Chapter 4

1. Find $\frac{dy}{dx}$ using implicit differentiation if
 - (1) $x^3 + y^3 = 3xy$.
 - (2) $\cos(xy^2) = y$.
 - (3) $\frac{1}{\sqrt{x}} + \frac{1}{\sqrt{y}} = 1$.

2. Find an equation for the tangent line to the graph of $x^4 + y^4 = 16$ at the point $(1, \sqrt[4]{15})$ using implicit differentiation.
3. Find the $\frac{dy}{dx}$ for the following functions:
 - (1) $y = \log_2(x^2 + x)$.
 - (2) $y = \frac{x^3}{1 + \log x}$.
 - (3) $y = x^2 e^{3x}$.
 - (4) $y = 3^{-x^2}$.
 - (5) $y = \ln\left(\frac{\sqrt{x} \cos x}{1 + x^2}\right)$. (Hint: using algebraic properties of the logarithm function)
 - (6) $y = e^{x \tan x}$.
 - (7) $y = \sin^{-1}(x^3)$.
4. Using the **logarithmic differentiation** to find the $\frac{dy}{dx}$ if
 - (1) $y = \frac{(x^2 - 8)^{1/4} \sqrt{x^2 + 1}}{x^5 + 4x + 3}$.
 - (2) $y = \sqrt[3]{\frac{x^3 - 1}{x^3 + 1}}$.
 - (3) $y = x^{\cos x}$.
 - (4) $y = (x^3 + 2)^{\ln x}$.
5. Determine whether the function f is one-to-one by examining $f'(x)$.
 - (1) $f(x) = x^8 + x + 1$.
 - (2) $f(x) = 2x + \sin x$.
6. Using L'Hopital rules to find the following limits:
 - (1) $\lim_{x \rightarrow 2} \frac{x^2 - 7x + 10}{x^2 - 25}$.
 - (2) $\lim_{x \rightarrow 0} \frac{\sin 4x}{x}$.
 - (3) $\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{\tan x}$.
 - (4) $\lim_{x \rightarrow +\infty} \frac{x^2}{e^x}$.
 - (5) $\lim_{x \rightarrow 0^+} (1 + \frac{2}{x})^{3x}$.